

Paper 1

Title	Author(s)	Description	Link
<i>VEHICLE: Validation and Exploration of the Hierarchical Integration of Conflict Event Data</i>	Benedikt Mayer, Kai Lawonn, Karsten Donnay, Bernhard Preim, and Monique Meuschke	A visual-analytics system for checking how conflict-event datasets from different institutions have been combined. Published in <i>Computer Graphics Forum</i> , 40(3), 2021.	Paper and DOI

You can read the local copy of the paper [here](#).

Context: Combining Conflict Datasets

Researchers often need to combine conflict records from different organizations. That isn't straightforward: the **same incident can appear in several databases** with different dates, locations, actor names, event categories, or levels of geographic detail. An incident may also appear in one source and be missing from another.

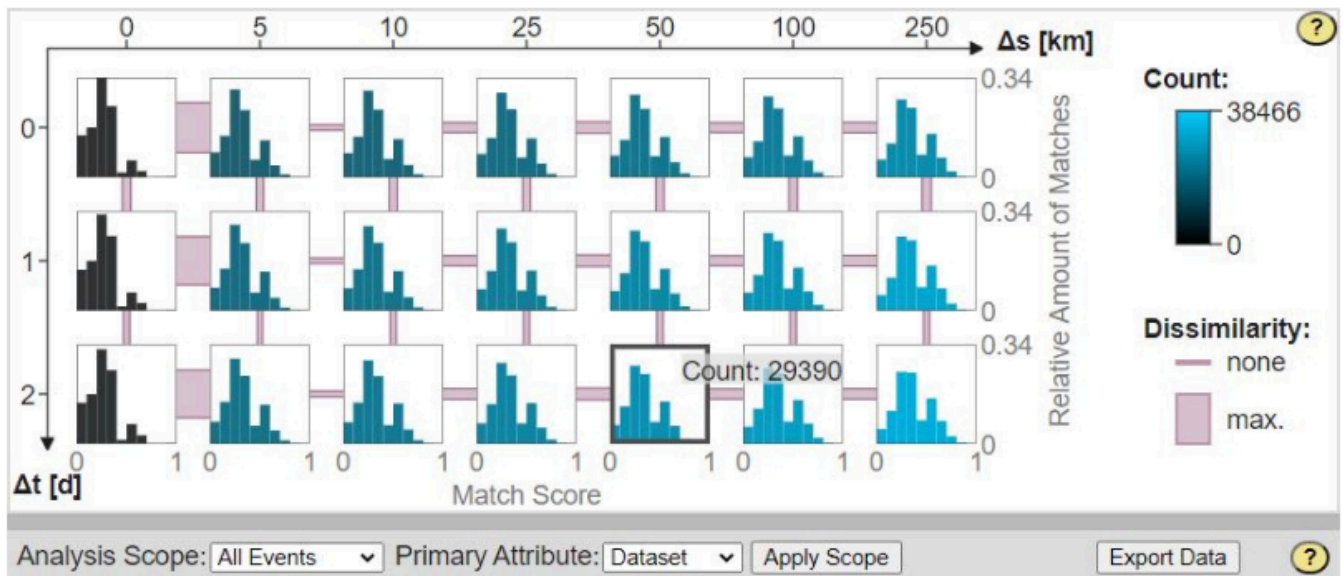
The paper looks at the results of MELTT, a semi-automatic procedure for matching and combining conflict datasets. MELTT **uses limits on distance and time, along with hierarchical classifications, to decide which records may describe the same event**. A table can list those matches, but it is harder to see the overall pattern.

Selected Representations

The interface uses **multiple coordinated views**. Each shows a different part of the matching problem, and they update together. This lets researchers examine whether two databases are describing the same events from several angles.

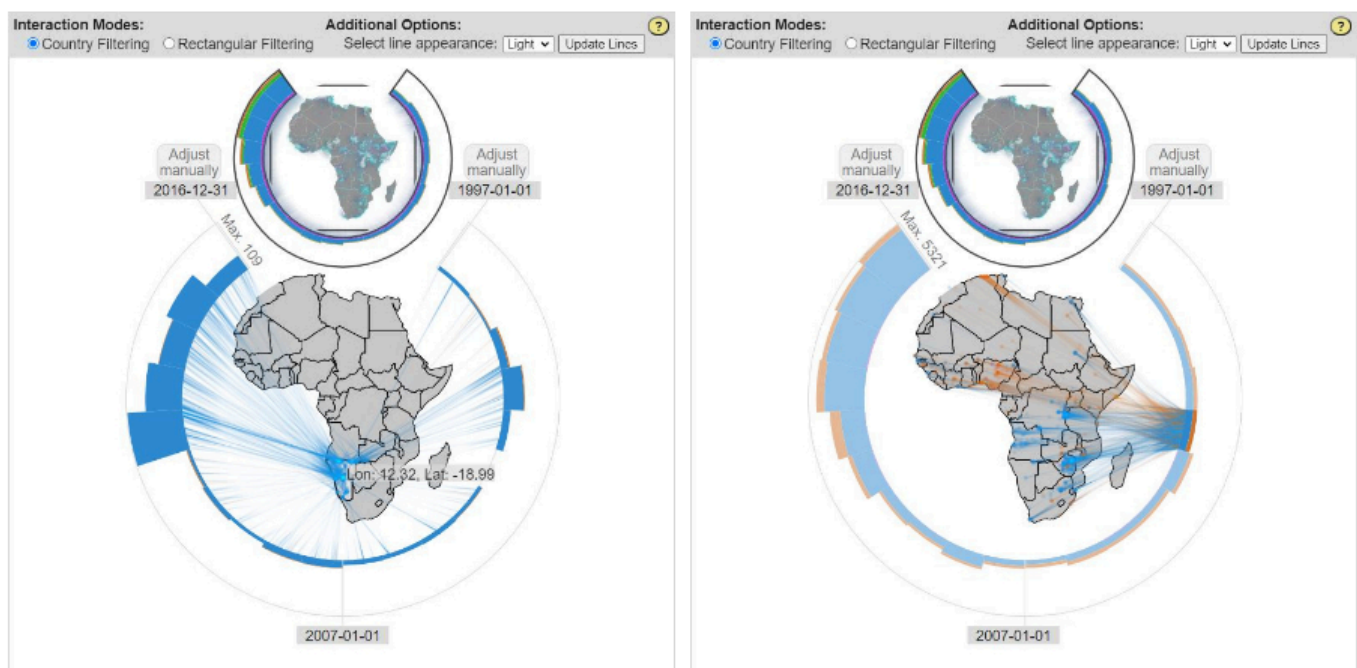
ParaMultiples

ParaMultiples shows a grid of small histograms. Each cell represents a combination of distance and time thresholds, and its histogram shows how many matches that setting produces and their quality. Researchers can compare nearby cells to see what changes when they adjust the settings.



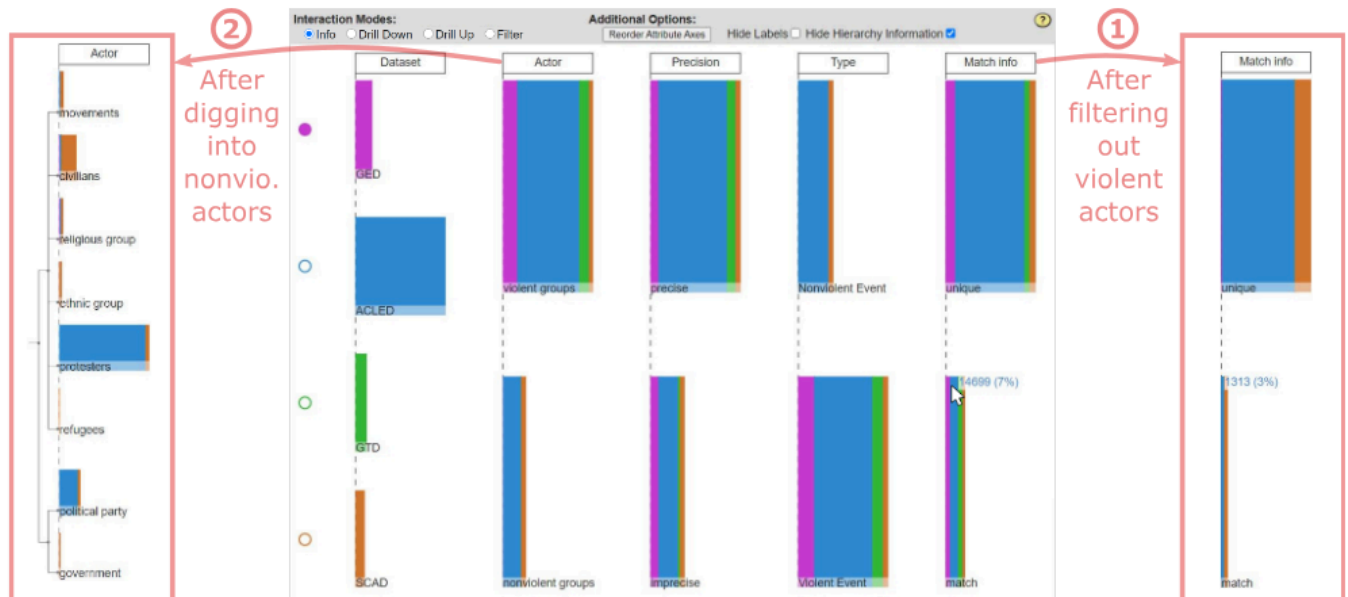
TempMap

TempMap combines a map with a radial time histogram. The map shows *where* candidate events happened, and the circular display shows *when*. Selecting events connects the two views, so researchers can look at place and time together.



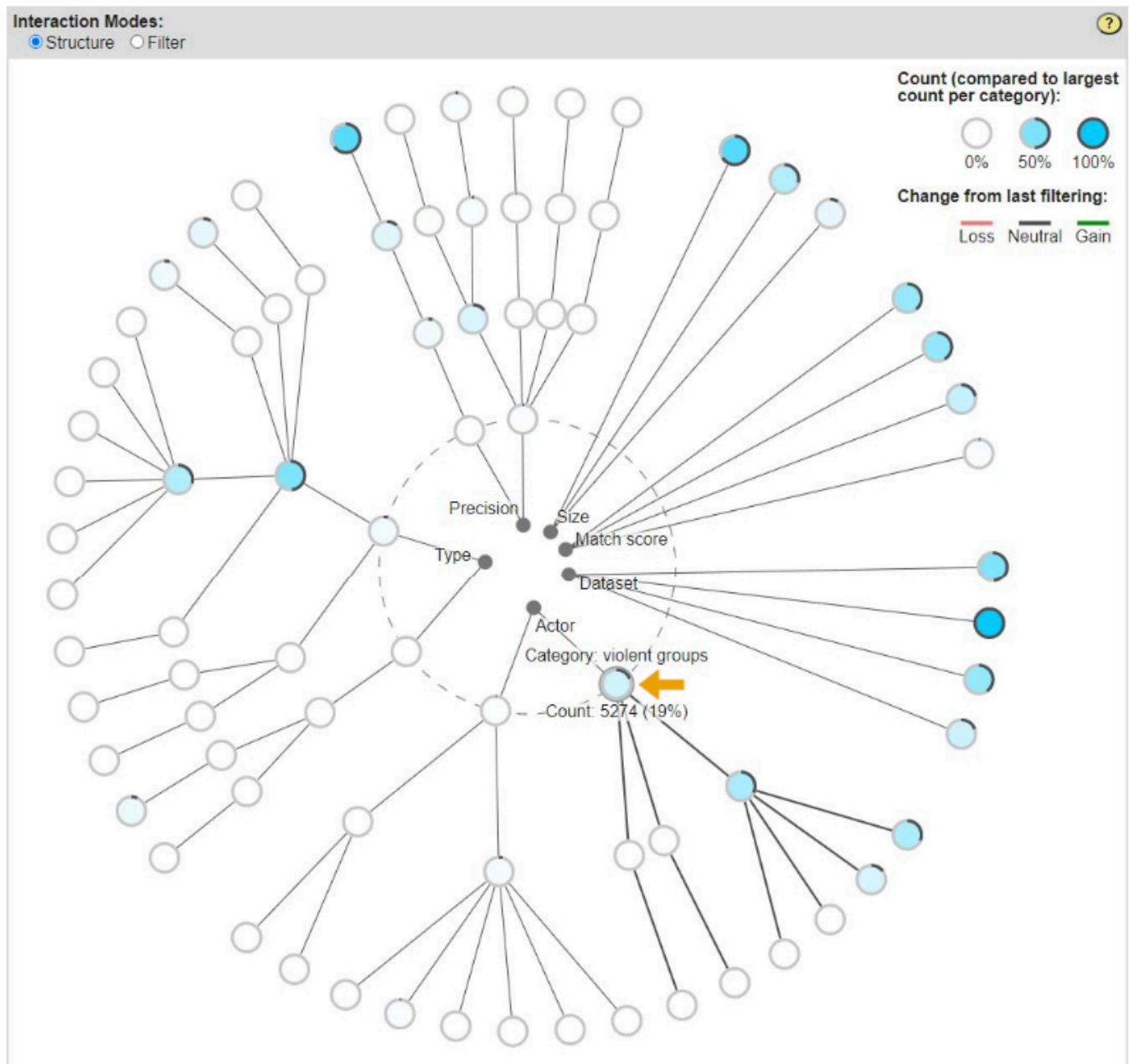
EventCharts

EventCharts use hierarchical stacked bars. Longer bars show larger amounts, segments show what makes up each total, and the hierarchy lets researchers move from broad event or actor classes to more specific ones.



MatchTree

MatchTree shows the matching classifications as a radial branching structure. Each branch belongs to a category, and the marks along it show where matches gather or disappear.



Other Options

Brushing, filtering, highlighting, and drill-down link the views. Select a bar, map region, parameter cell, or tree branch, and the same records are highlighted across the interface.

This makes it easier to follow a problem through the different views.

Knowledge: What can we learn from it?

The linked views help show whether agreement and disagreement between sources follow **systematic patterns**. A table, instead, can only tell us which records matched.

The interface also shows whether matching failures gather in a particular country, period or part of the event classification. For example,

- Clusters in TempMap may point to uneven geographic or time coverage, while

- Uneven stacked bars may suggest that one source supplies a disproportionate share of certain event types.

In particular ParaMultiples helps researchers judge **how stable the integration is**. If the nearby cells look similar, small changes to the thresholds probably make little difference to the combined dataset. On the other hand, a sudden change between cells points to a parameter "cliff": a small choice produces a very different result. **That kind of sensitivity is almost impossible to spot in a match table.**

The views also **make it easier to notice local problems in results that look good overall**. A setting may produce an acceptable total while the map or classification reveals a serious failure.

Thus, we can learn *where the sources agree, where they disagree, which choices affect that disagreement, and whether the combined dataset is reliable enough for a particular research question.*

Pipeline from raw information to the visualizations

From the paper, we can briefly summarize the representations pipeline as follows:

1. Data is collected autonomously by ACLED, UCDP GED, the Global Terrorism Database, and the Social Conflict Analysis Database.
2. The records are standardized and sorted using shared classifications for actors, event types, and geographic precision.
3. MELTT uses distance and time thresholds to find possible matches, then calculates similarity using the classifications.
4. VEHICLE calculates counts, distributions, match scores, and differences between settings.
5. The results appear in linked maps, histograms, hierarchical bars, and radial trees.

Sources and tools

- **Primary sources:** [ACLED](#), [UCDP GED](#), [Global Terrorism Database](#), and [Social Conflict Analysis Database](#). The study covers 197,502 conflict events in Africa from 1997 to 2016.

Paper 2

Title	Author(s)	Description	Link
<i>A Visual Analytics Framework for Identifying Topic Drivers in Media Events</i>	Yafeng Lu, Hong Wang, Steven T. Landis, and Ross Maciejewski	A visual-analytics framework for exploring possible links between media attention and outside event drivers, using media collections and an armed-conflict event dataset. Published in <i>IEEE Transactions on Visualization and Computer Graphics</i> , 24(9), 2501-2515, 2018.	DOI · Institutional repository

You can read the local copy of the paper [here](#).

Context: Words Matter

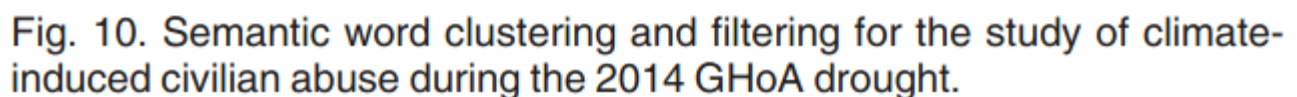
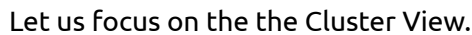
This paper presents a visual-analytics framework for **exploring whether topics in media collections relate to possible outside event drivers**.

Researchers can retrieve events through related words, test possible causal links, and add annotations. They can also inspect the words behind a match and use their knowledge of the subject to refine it.

Here, we look at **Figure 10 in Section 7.2**, "Climate-Induced Unrest During Drought." The case study asks whether the 2014 drought in the Greater Horn of Africa coincided with reports of social unrest and political violence. The analyst selects an agriculture topic from a social-unrest media collection, ACLED events labelled "violence against civilians," and the starting terms **water**, **food**, **farmer**, and **climate**.

Representation: Cluster View

For context we show the full analytics interface



More in detail,

- **Each circle is a word**, and filled circles are the selected media keywords. Plus, outlined circles are related words found in ACLED event descriptions by the semantic model.
 - Colour shows the starting search concept: **farmer** is blue, **climate** red, **water** tan, and **food** orange.
 - **Circle size is proportional to how often a word appears in the event records**. Large nodes such as *fire*, *air*, *water*, and *food* draw more attention than small terms around the edges.
 - **Distance and outlines show groups of related words**. Similar words attract one another, and automatically drawn boundaries separate the groups formed by the similarity threshold and the analyst's changes.
 - The dashed box on the right is the **word-selection area**. Words or smaller groups moved into it become the filters used to retrieve events.
-

Knowledge: What can we learn from it?

A table could list the keywords and their similarity scores. The bubble groups make **ambiguity in word meaning** easier to see, showing how one search term can lead in several directions. Furthermore, moving chosen words into the selection box changes the query, so the analyst's reading of the graphic directly affects which events are retrieved.

This tells us the same word can mean different things in different datasets. In fact, seeing the associations **helps an expert understand why the retrieved events might be misleading.**

Pipeline from raw information to the visualizations

Taking the example of the paper to describe the phases, we can say that:

1. The analyst selects the topic from the data collections and the labels. This analysis focuses on the months around the 2014 Greater Horn of Africa drought.
2. The words *water*, *food*, *farmer*, and *climate* provide the starting points for a semantic dictionary. Then, the text is normalized, and words with related meanings are extracted from ACLED event descriptions.
3. Here, we have a complete-link agglomerative clustering and a force-directed layout to group similar words. Colour shows the starting keyword, and circle size shows frequency in event text.
4. The analyst adjusts the similarity threshold, moves words between groups, puts accepted terms in the selection box, and removes misleading associations such as the *climate* group.
5. The filtered events are grouped into a time series and passed to the causality model to test the hypothesis.

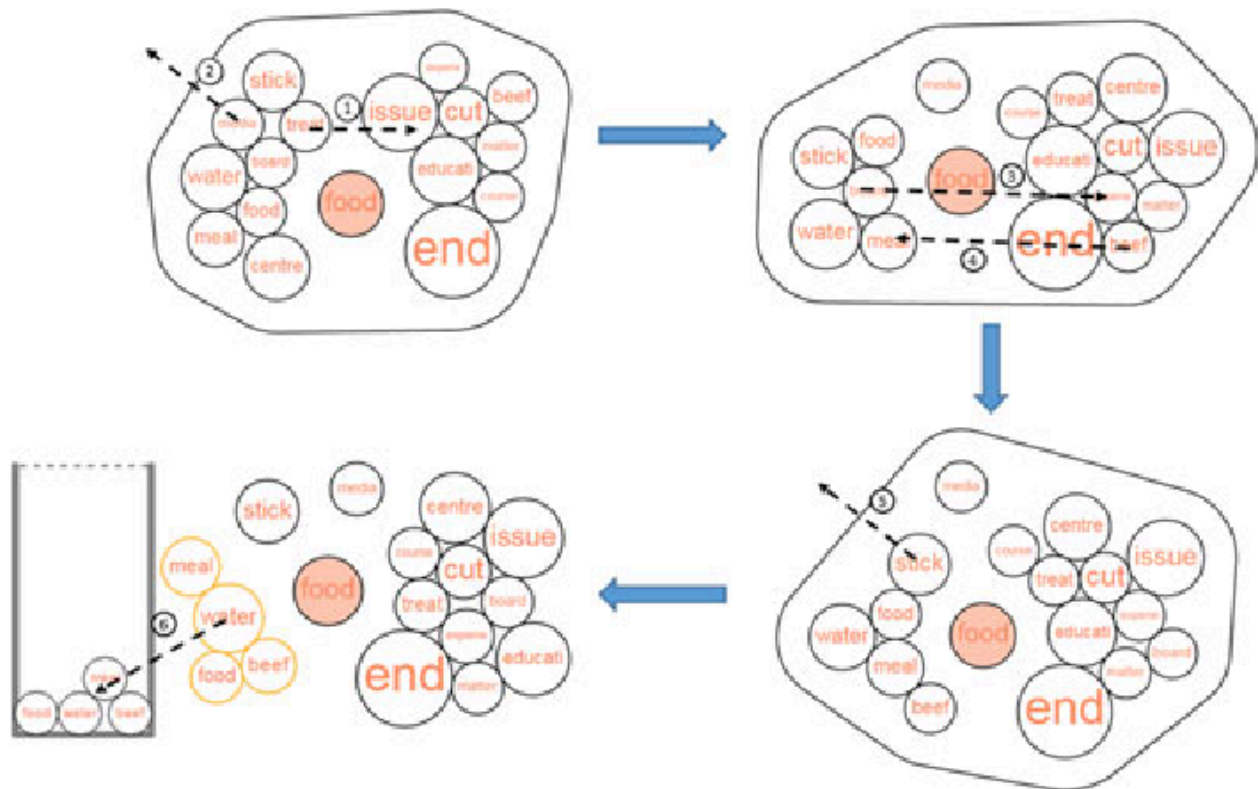


Fig. 4. This figure shows an interaction process of steering the cluster force layout to update the word concept clusters. Each step is marked numerically on the figure. User can drag a word away from all clusters or towards another cluster. User can drag a group of words to select them.

Sources and tools

- **Primary sources:** [ACLED](#), [The Climate Change Media Dataset](#), [Social-Unrest Media Collection](#).

Responsive Web Article 1

Title	Author(s)	Description	Link
<i>How Drone Combat in Ukraine Is Changing Warfare</i>	Reuters Graphics; Mariano Zafra, Max Hunder, Anurag Rao, Sudev Kiyada, and collaborators	A visual investigation that works on desktop and mobile, explaining drone roles, operation, range, cost, and how they work together on the battlefield in Ukraine.	Reuters article

Context: Drone Combat

When the article was published in March 2024, drones had become central to direct attacks and long-range strikes in Ukraine.

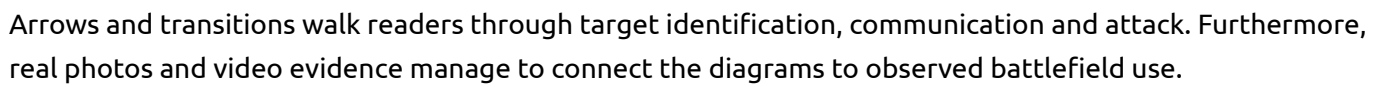
Reuters needed to explain how this technology works on the battlefield to a broad audience, including citizens, journalists, professionals, and people interested in policy.

Representations: Drone and Maps

The article uses **scroll-driven visual explanation**. As readers scroll it can see routes, ranges, and tactical stages appear.

The story (i) starts with an individual drone, (ii) then moves to the nearby battlefield, and finally (iii) it shows the wider geography of long-range strikes.

The drone scene stays in place while details change, making it easier to focus on one relationship at a time. Later on, there are also labelled drawings explain drone components and roles. Maps and distance comparisons show how far drones can travel, while silhouettes and proportional drawings show their physical size.



This last map gives a context to "numbers" such as range and cost. On mobile, wide scenes become vertical sequences, with labels reduced or moved so the explanation stays in the same order.

Overall, the article makes readers follow an explanation chosen by the journalists. There is a storyline that connects the zoom-in of the drone's components to a zoomed-out map of the whole conflict. The reading and understanding is facilitated by short passages, animations, and responsive graphics.

Knowledge: What can we learn from it?

A table could compare drone models, costs, speeds and ranges. The diagrams help us see **how their roles depend on one another**:

1. Reconnaissance drones find the activity,
2. Communications pass on the information,
3. Artillery or attack drones respond,
4. Longer-range systems reach beyond the front.

Their **battlefield effect depends on how these roles work together**.

The **changes in scale also help us understand distance**. "Ten kilometres" or "several hundred kilometres" becomes a **visible relationship between the front line**, cities and target areas.

Putting observation, decision, and attack in one sequence shows **how low-cost drone systems can immediately act after spotting activity**. It also shows the imbalance between relatively cheap devices and potentially high-value targets. These relationships would be hard to pick up from a table.

Pipeline from raw information to the visualizations

Unfortunately we initially had **no description** of how the visualization was made.

Sources and tools

- **Primary sources:** [Royal United Services Institute \(RUSI\)](#), [ACLED: Armed Conflict Location & Event Data Project](#) (Drone attacks from Feb. 24, 2022, to March 12, 2024), [Institute for the Study of War](#) with [American Enterprise Institute's Critical Threats Project](#) (Russian-occupied territory data as of 3 p.m. EST, March 12), [Council on Foreign Relations](#), [CSET: Center for Security and Emerging Technology](#)

For what concerns the videos of FPV drone attacks, the article states that:

"The videos have been verified by Eleanor Whalley, of Reuters' Visual Verification team. They were sourced from Telegram channels or provided by various drone units of the Ukrainian Army (including the Azov Brigade in Kremina, the 110th Mechanised Brigade in Lastochkyne and Avdiivka, and the State Border Service of Ukraine in Vodyane and Hryanykivka)."

Responsive Web Article 2

Title	Author(s)	Description	Link
<i>How Ukraine Has Faced Its Worst Month on the Battlefield in Two Years - Visualised</i>	Ed Gargan, Pablo Gutiérrez, and Ashley Kirk; design by Prina Shah	A series of charts and maps for desktop and mobile showing Russian territorial gains in Ukraine in 2024, with a focus on November.	The Guardian article

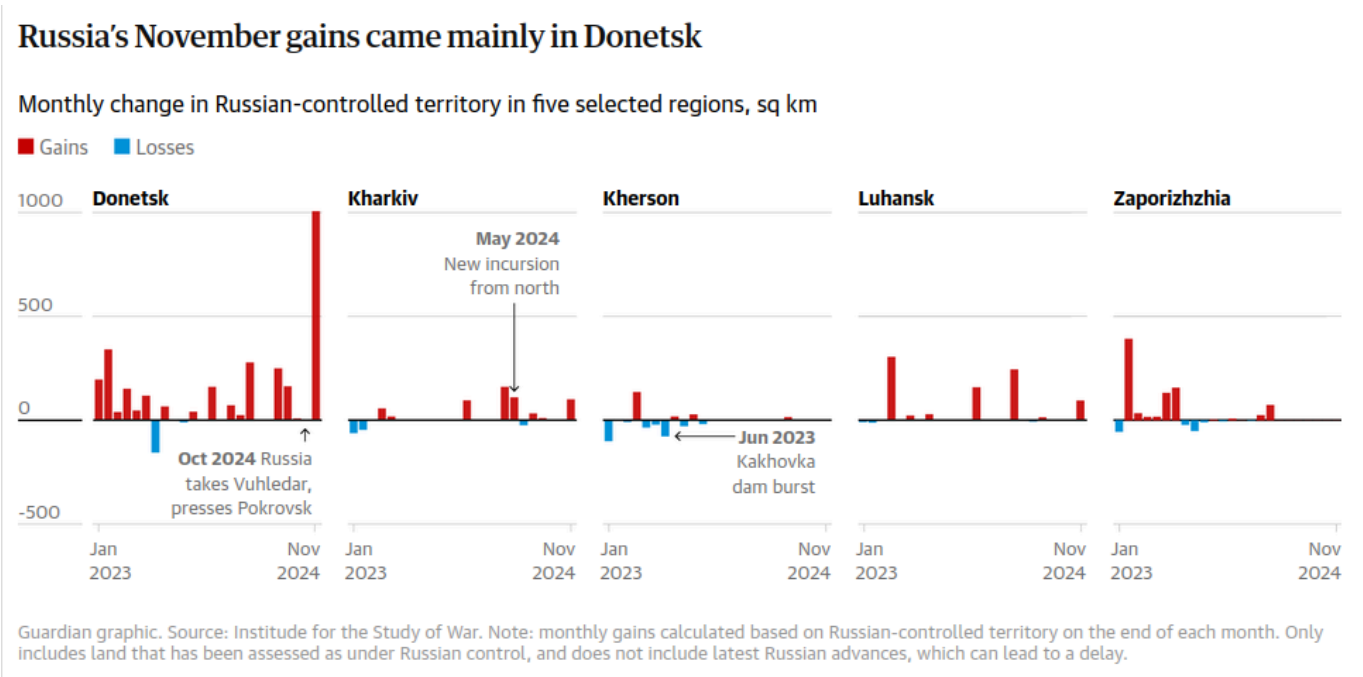
Context: Lost Terrain

The article was published in December 2024, after Ukraine lost more territory in November than in any month since September 2022. The authors look at how those losses compare with earlier months, where they happened, and what the captured land contained, **so readers can judge what the area totals mean strategically**.

The main source is the Institute for the Study of War's daily control-of-terrain files. These distinguish areas under control, contested areas, and areas where advances have been recorded.

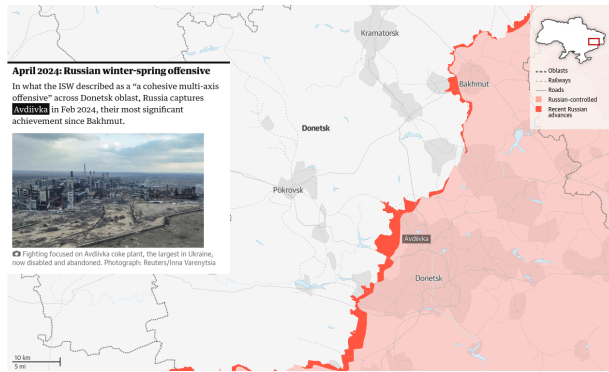
Representations: Lost Terrain and Chart

The article combines a **time series of territorial change** with **annotated control-of-terrain maps**. The chart lines up the months on one time axis and shows the amount of land gained or lost. This makes faster changes and unusually large monthly losses easy to spot. A note highlights November 2024.

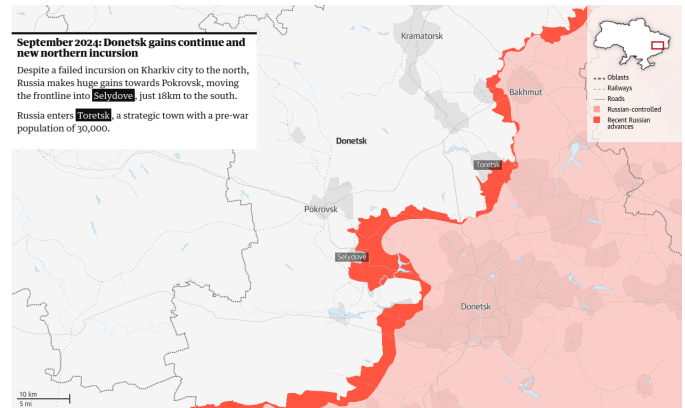


The maps show what those square kilometres look like on the ground. Colour distinguishes control, advances, and contested areas.

April 2024



September 2024



Regional boundaries locate the changes, while settlement labels and infrastructure references **help explain their strategic importance**. The story moves from Ukraine as a whole to Donetsk and, then, to individual advances, connecting the national totals to local places.

The order of the story connects the two views. First, the time series shows *when the change became exceptional*. Then, the maps explain *where it happened and what was in the captured area*.

Overall, the article gives citizens and professionals a clear view of a changing battlefield. It manages to give a simple representation of the changes in daily control files while keeping the difference between area gained and strategic importance visible. As for UX, desktop readers get more detail, and mobile readers can follow the charts and maps in a vertical sequence.

Knowledge: What can we learn from it?

A table can tell us that Russia captured approximately 1,202 square kilometres in November 2024. The **chart shows why that number matters**: it stands out from the surrounding months and marks Ukraine's worst monthly territorial loss in roughly two years.

The **maps add context to that total**. Most of the change happened in Donetsk, and much of the area was agricultural land, forest, or fields rather than major cities or transport hubs. We can see that **the amount of land gained and its strategic value are related, but they are not the same thing**.

The shape of the advances also tells us how the losses happened. The 1,202 square kilometres were gained through several gradual, uneven movements.

Looking at the chart and maps together, we can see that November brought severe losses in area, while the consequences depended many factors such as distance from hotspots.

Pipeline from raw information to the visualizations

Unfortunately we have **no description** of how the visualization was made. However, the claims the property of the graphics and we have a few details in the images' descriptions:

- For the barchart: "Monthly gains calculated based on Russian-controlled territory on the end of each month. Only includes land that has been assessed as under Russian control, and does not include latest Russian advances, which can lead to a delay."

- For the other one: "Only includes land that has been assessed as under Russian control, and does not include latest Russian advances."

Sources and tools

- **Primary sources:** Daily control-of-terrain files from the [Institute for The Study of War](#).

Instagram Post 1

Title	Author(s)	Description	Link
<i>The 50 Countries Most Impacted by Violent Conflict</i>	Powerful Countries; data source: ACLED Conflict Index	An Instagram map based on the ACLED Conflict Index, showing the fifty countries and territories most affected by violent conflict.	Instagram post

Context: Political Violence

ACLED records political violence with details such as dates, locations, actors, event types, and fatalities. The Conflict Index brings these records together into four dimensions: deadliness, danger to civilians, geographic diffusion, and armed-group fragmentation.

The Instagram account Powerful Countries turns that detailed analysis into a **world overview that people can take in quickly while scrolling their feed**.

Representation: Top 50 Countries by Violence

POWERFULCOUNTRIES

50 COUNTRIES WITH HIGHEST RATES OF VIOLENT CONFLICT

The ACLED Conflict Index calculates where conflicts in every country and territory in the world vary according to four indicators – deadliness, danger to civilians, geographic diffusion, and the number of armed groups. Using political violence event data of the past twelve months, the ACLED index uses the four indicators of deadliness, danger, diffusion and group fragmentation to identify the country or territory with the highest score

EXTREME HIGH TURBULENT/LOW

PowerfulCountries
An Exeter Brand

Geography Can Tell Any Story

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Here is a map showcasing the top 50 countries most impacted by violent conflict, according to the ACLED Conflict Index.

The ACLED Conflict Index calculates where conflicts in every country and territory in the world vary according to four indicators – deadliness, danger to civilians, geographic diffusion, and the number of armed groups.

Using political violence event data of the past twelve months, the ACLED index uses the four indicators of deadliness, danger, diffusion and group fragmentation to identify the country or territory with the highest score on the individual indicators, as well as the top 50 countries and territories experiencing extreme, high, or turbulent levels of conflict as a combined score on the four indicators.

919 100

July 17, 2025

Log in to like or comment.

The main image is a **categorical world map**.

1 / 3

Each country stays in its geographic position, while **colour and emphasis identify the fifty countries** and territories included in the Conflict Index and distinguish their **conflict severity**. Instead, the remaining states are left blank, which helps the highlighted regions stand out on a phone screen.

The main feature of this representation make the map quick to read:

- Small set of colours
- Compact legend
- Clear headline
- Few labels

The only limit is given by the fact that, trivially, country size determines how much space it takes up. As a consequence, a large country can therefore draw more attention even if its index value is not proportionally higher (lower density of violence). Furthermore, the huge title would not be a problem if it was not accompanied by a long description inside the picture. This is justified from the map not being able to be displayed in a larger format.

The description clearly includes the sources, a brief description on the data's time range and the criteria for coloring.

Knowledge: What can we learn from it?

The map helps us see **where affected countries cluster and which ones are neighbours**. In fact, a ranked table shows each country as a separate entry. Here, instead, regional groups and connected border zones become visible. The highlighted and blank areas also show how unevenly conflict burden is spread around the world.

These patterns invite to question about what nearby conflicts may share. Plus, the diagonally oriented distribution of the colors seems to cut the world in half. In a way, it points to the fact that we must ignore these conflicts, because they are closer than they look.

One problem that persist, it that the map helps us spot patterns, but it cannot tell us why two countries received a similar colour. For that, we need the four index components.

To sum up, this visualization let us learn about where severe conflicts cluster within regions and across borders, and how similar classifications can reflect different measures of severity.

Pipeline from raw information to the visualizations

Luckily, we find some useful information in the post's description:

1. They took the data from ACLED, which codes events by date, location, event type, actors, fatalities, and civilian targeting. The events cover around 12 months from July 2024 to July 2025 from the description states.
2. The Conflict Index methodology groups these events together and all the countries are assigned a label directly related to the amount of total violence.
3. The top fifty countries are selected and shown on a simplified categorical world map, along with the label assigned.

Sources and tools

- **Primary sources:** [ACLED Conflict Index](#).

Instagram Post 2

Title	Author(s)	Description	Link
25 Years On: The Human Toll of 9/11	Forbes India / Network18 Creative	A comparison of estimated deaths in post-9/11 wars, the Vietnam War, the Korean War, and the 1991 Gulf War, marking the 25th anniversary of the September 11 attacks.	Instagram post

Context: War Comparison

War-death estimates usually come in reports and tables but, for a general reader, a **table doesn't immediately show how the scale of one war compares with another**.

Forbes India published this post, on 11 September 2026, which compares estimated deaths in major wars involving the United States since 1945.

Representation: Human Cost

25 YEARS ON:
THE HUMAN TOLL OF 9/11

The attacks of September 11, 2001 triggered wars whose human cost would ultimately run into millions

Estimated death toll in major US-involved wars since 1945

4.5–4.7 million	~3 million+	~2.5–3 million	~143,000–206,000
Post-9/11 wars*	Vietnam War	Korean War	1991 Gulf War

*INCLUDES DIRECT DEATHS FROM VIOLENCE AND INDIRECT DEATHS FROM WAR-RELATED DISEASE, MALNUTRITION AND INFRASTRUCTURE COLLAPSE

SOURCES: BROWN UNIVERSITY COSTS OF WAR PROJECT; US GOV'T DATA; REUTERS

Network 18 creative

DAILY DATA

THE TOLL CONTINUES

Thousands of responders and survivors have since died from illnesses linked to 9/11, including more than 9,000 deceased members of the World Trade Center Health Program

Forbes

forbesindia

Follow

forbesindia 6d

A look at the human cost of 9/11 and the wars that followed, 25 years on.

barakzilberberg 6d

America 🇺🇸 stands strong today as a resilient nation, carrying forward the memory and legacy of the heroes and lives lost 25 years ago on September 11, 2001. Across the country, communities are coming together for the 25th anniversary of 9/11 to honor the enduring spirit, bravery, and unity of the American people.

Like Reply

mohmed_azhr 6d

After 9/11, the United States launched the "War on Terror" and carried out

93 2 6 days ago

Log in to like or comment.

First of all, the graphic places **four estimated death ranges** alongside one another in a horizontal layout:

- **Post-9/11 wars:** 4.5 – 4.7 million;
- **Vietnam War:** approximately 3 million or more;
- **Korean War:** approximately 2.5 – 3 million;

1 / 3

- **1991 Gulf War:** approximately 143,000 – 206,000.

Each of the estimates comes with a block of repeated **human pictograms** and documentary photograph (which provides context). However, it is not indicated whether there is a fixed value (in terms of lost human lives) for each icon.

Another aspect to consider is that, the blocks **differ in size** (directly proportional to the estimates), which is used to sort them from left to right in descending order. On the other hand, grouping multiple wars for the "Post 9/11" wars feels like cheating if they do not mention which wars they are referring to (ex: Iraq).

A **side panel** covers illnesses among 9/11 responders and survivors, including more than 9,000 deceased members of the World Trade Center Health Program.

The layout works well for a phone screen: the title introduces the point, the numbers keep the ranges visible, and the blocks of figures give a quick sense of scale. Unfortunately, the image seems to be a bit packed.

The **usage of the red/grey contrast draws attention** to post-9/11 deaths before the historical comparisons.

Overall, elements such as a tragic anniversary, human figures, contrasting colours, and a few comparisons **make numbers easier to understand within seconds**.

Knowledge: What can we learn from it?

The benefits of this visualizations are multiple.

First, the **blocks make the differences in scale really easy to see**. The reader is brought to think that, due to the size, Post-9/11 wars form the largest group. In other words, this conveys the message that active conflicts have not stopped in modern times and that they are even more fatal.

Secondly, **the graphic includes deaths that a battle-death table might leave out**. We know well that bombs and bullets are not the only aspect of war that brings death. In fact, damaged healthcare, hunger, diseases, and infrastructure collapse are all fatal, but they are not always taken into account. This comparison helps us focus on a new definition of war fatalities, one that recognizes how the consequences of destruction affect people's lives (even after the end of a conflict).

Unfortunately some aspects like, grouping multiple wars estimates and comparing different historical periods (with different world demographics) might not seem fair since we might not have sources for non-combat deaths.

Pipeline from raw information to the visualizations

Unfortunately we have **no description** of how the visualization was made. However, from the image we infer that the possible transformations done include: choosing conflicts, combining direct and indirect deaths, simplifying ranges, and scaling the pictograms.

Sources and tools

- **Primary sources:** [Brown University Costs of War](#), US government data, and Reuters.

X Post 1

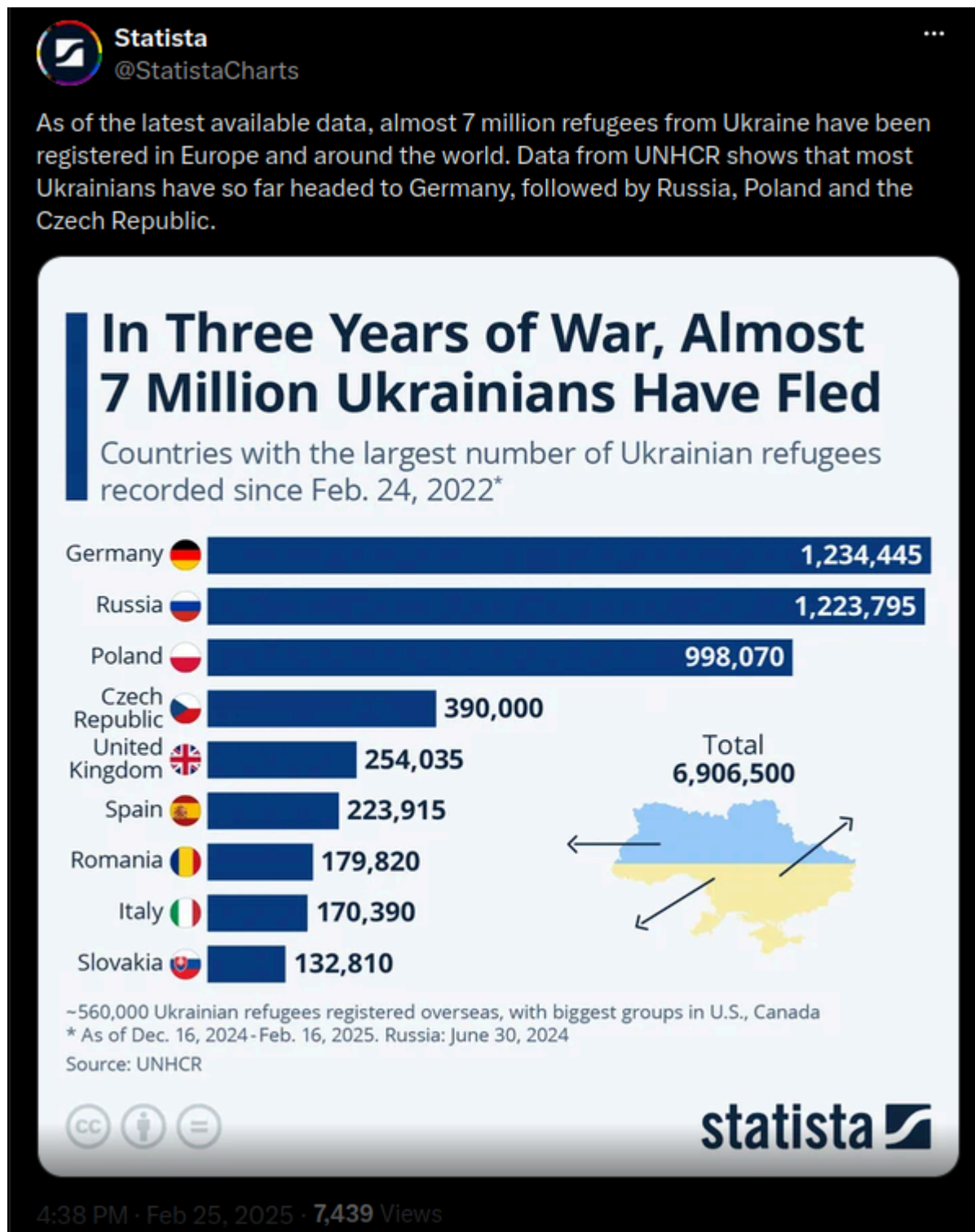
Title	Author(s)	Description	Link
<i>In Three Years of War, Almost 7 Million Ukrainians Have Fled</i>	Statista	An X post with a horizontal bar chart comparing the countries with the most registered refugees from Ukraine, using UNHCR data.	Post on X

Context: War Refugees

Russia's invasion of Ukraine forced large numbers of people to cross borders. UNHCR records refugee populations by host country and reporting date. Statista uses those records for a compact ranking on X, letting readers compare countries quickly while retaining the exact figures found in the tables.

The post was published on 25 February 2025. It reports almost 7 million registered refugees in Europe and elsewhere and highlights the countries with the largest registered populations.

Representation: Ukranian Refugees in Europe



The attached image is a **ranked horizontal bar chart**.

The title is the first thing that catches the eye, due to its size, its bold font and the fact it shows the horrible consequence of war.

Country names and flags line up on the left, and all bars use the same numerical scale. Bar length shows the number of registered refugees, end labels give exact values, and ordering the bars from largest to smallest makes the ranking easy to follow. Furthermore, limiting to top 7 countries, helps the bars fit the image.

The horizontal layout leaves room for readable country names and large values while **fitting the ranking into one mobile image**.

A short source and date note at the bottom tells readers where the figures came from.

Last but not least, the representation also includes blue and yellow silhouette of Ukraine, with some arrows representing the main exit routes.

Overall, the post fits how people use X: it is topical, quick to read, and easy to share. Here, a large dataset becomes an answer to the question "where were refugees registered?". Readers can see the pattern without opening another page, and the source and date notes give them a way back to the evidence.

Knowledge: What can we learn from it?

The bars make the **shape and tiers** of the distribution easy to see. Germany, Russia, and Poland each have roughly one million or more registrations. There is then a sharp drop to the Czech Republic, followed by much smaller totals for the other displayed countries. Of course, these gaps are easier to spot in bar lengths than in a column of numbers.

The overall total could hide how uneven the distribution is. A few countries account for a large share of the registrations shown. This raises questions about how the counts compare with host populations and how they have changed over time, neither of which this absolute-count chart answers.

The footnote matters here, since the country figures do not all refer to the same date.

Pipeline from raw information to the visualizations

Unfortunately we have **no description** of how the visualization was made. However, from the image we infer that the possible transformations done include:

1. UNHCR country-level records are selected for refugees displaced from Ukraine since 24 February 2022.
2. Countries are ranked by the number of registered refugees.
3. The largest destinations are shown as aligned horizontal bars. Exact values, an overall total, and a short source note make the image understandable on its own.

Sources and tools

- **Primary sources:** The [UNHCR Ukraine Refugee Situation portal](#) for people displaced from Ukraine.

X Post 2

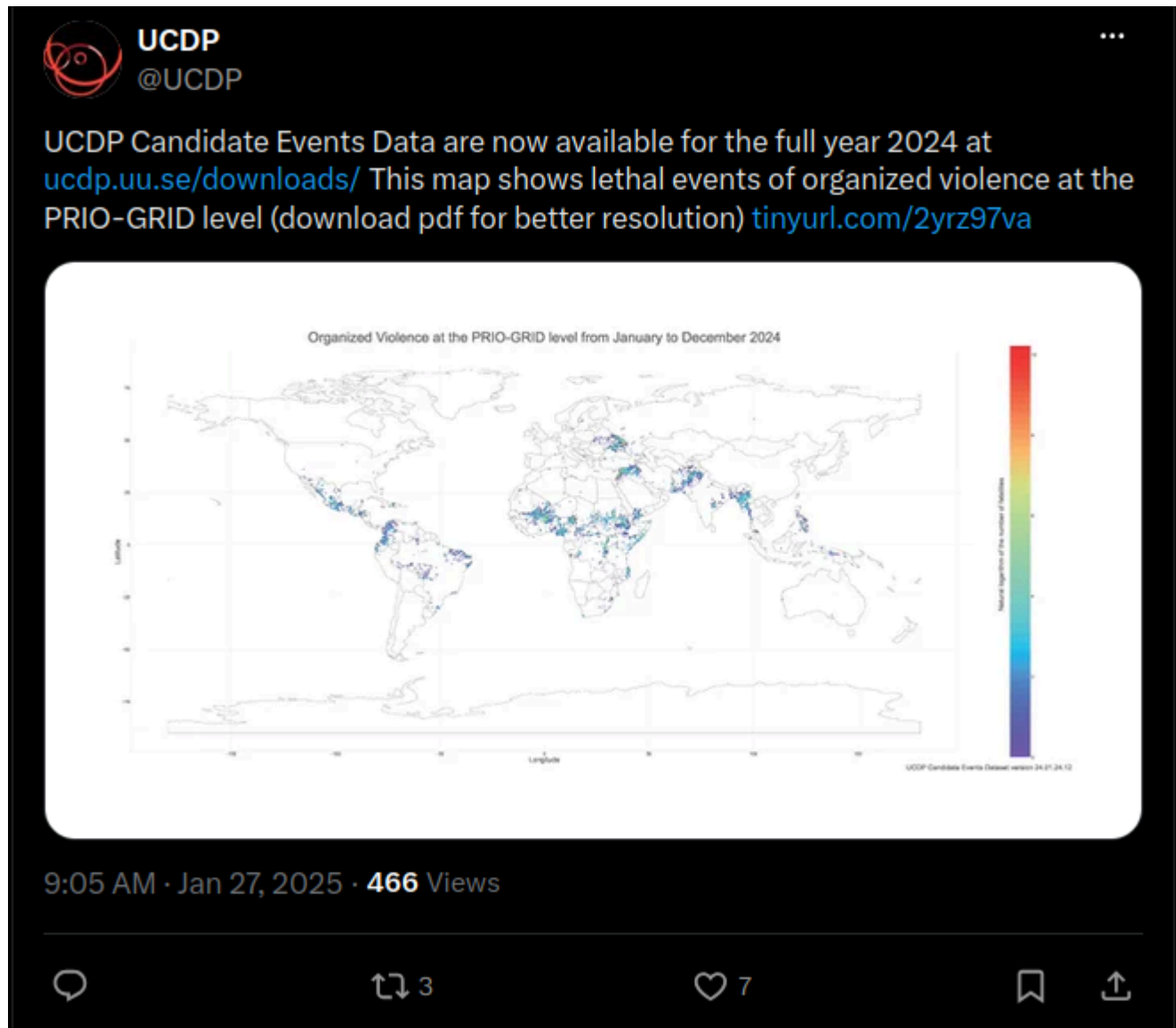
Title	Author(s)	Description	Link
<i>UCDP Candidate Events Data: Lethal Events at PRIO-GRID Level</i>	Uppsala Conflict Data Program	An X post announcing updated candidate-event data and showing lethal organized violence on a standardized spatial grid.	Post on X

Context: Events of Violence

UCDP Candidate Events provide near-real-time records that have not yet gone through the full annual release process. There are too many records to show individually in a post, and a global total would lose the geographic pattern.

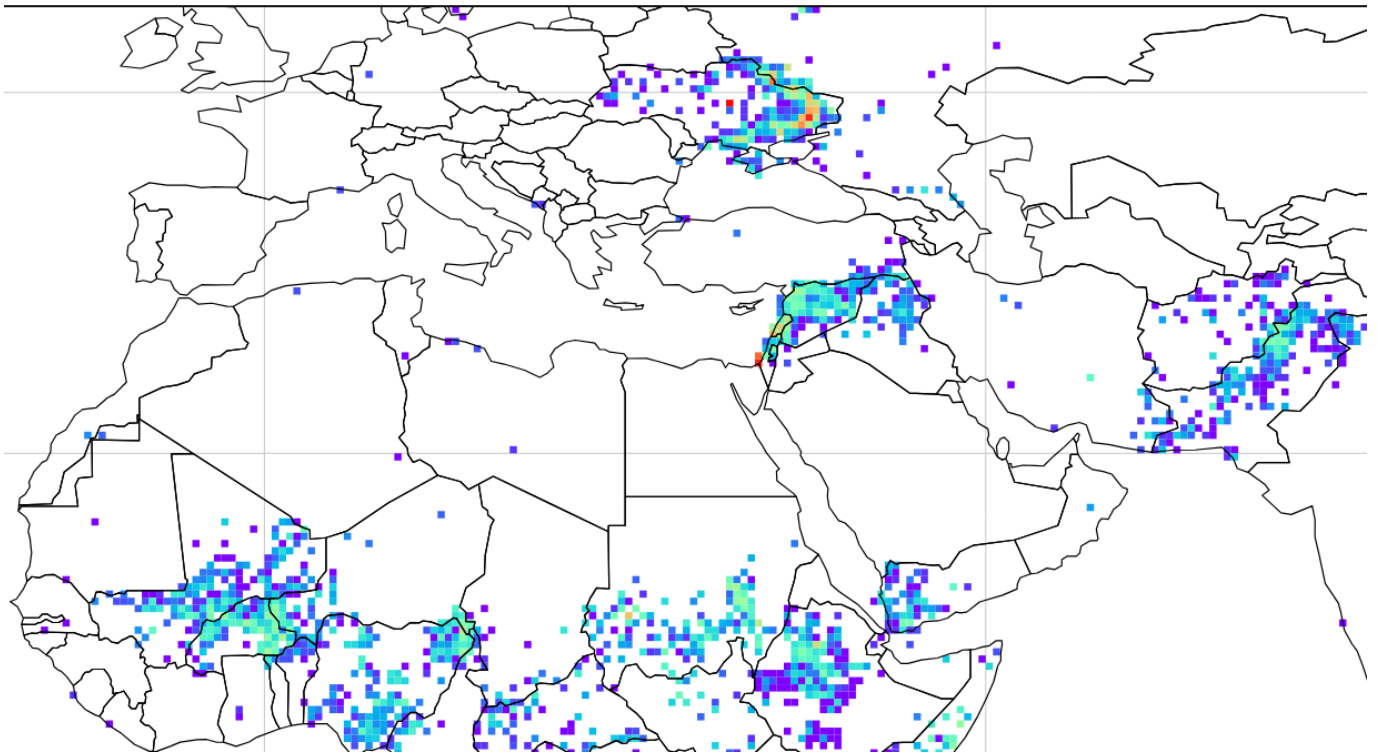
The post groups lethal events into PRIO-GRID cells, using the same spatial unit around the world to give readers a quick map update.

Representation: Lethal Events of Organized Violence



The image is a **gridded event map**.

Each event with a location is assigned to a PRIO-GRID cell. **These cells provide a common spatial unit**, instead of grouping events by some aspect.



Filled cells or changes in intensity show where lethal organized violence was recorded during the stated period. The value used for this intensity is the natural logarithm of the numbers of fatalities. This was probably a choice to avoid extreme values (as information theory teaches us).

Multiple active cells in a close area tend to form clusters or even corridors, which can be useful to draw violence boundaries or study patterns.

The simple legend and short notes keep the image readable in the X feed.

Overall, the post reads like a research announcement: current, concise, and clear about its source. That gives it a different tone from the selected Instagram examples.

Knowledge: What can we learn from it?

The map shows **where violence gathers, spreads, and connects across space**. While a table of coordinates identifies individual events, here, adjacent cells help us see whether lethal violence forms

- One dense hotspot,
- Separate conflict areas,
- Or a cluster crossing a border

The grid does not match the actual borders. **That helps us notice cross-border connections and conflict zones near the edges of countries or provinces**. It is especially useful where armed conflict follows certain routes or ethnic regions.

Unfortunately, the grouping also stops us from learning, for example, who was responsible or how uncertain the original coordinates might be. It helps us understand *where violence clusters*, but leaves questions about *who acted, why, and what happened in each event* open.

Pipeline from raw information to the visualizations

Unfortunately we have **no description** of how the visualization was made.

Sources and tools

- **Primary source:** The UCDP Candidate Events Dataset for 2024.

Alternative Paradigm Representation 1

Title	Author(s)	Description	Link
<i>Russia Bombs Ukraine Almost Every Night. This Is What It Sounds Like.</i>	<i>The Washington Post</i> : Lizzie Johnson, Serhii Korolchuk, Anastacia Galouchka, Kostiantyn Khudov, Ed Ram, Yutao Chen, Júlia Ledur, Bishop Sand, and collaborators	A 2025 interactive story about nights of aerial war, combining recordings, scroll-driven audio, drone-launch charts, loudness comparisons, and air-raid-siren durations.	Interactive article

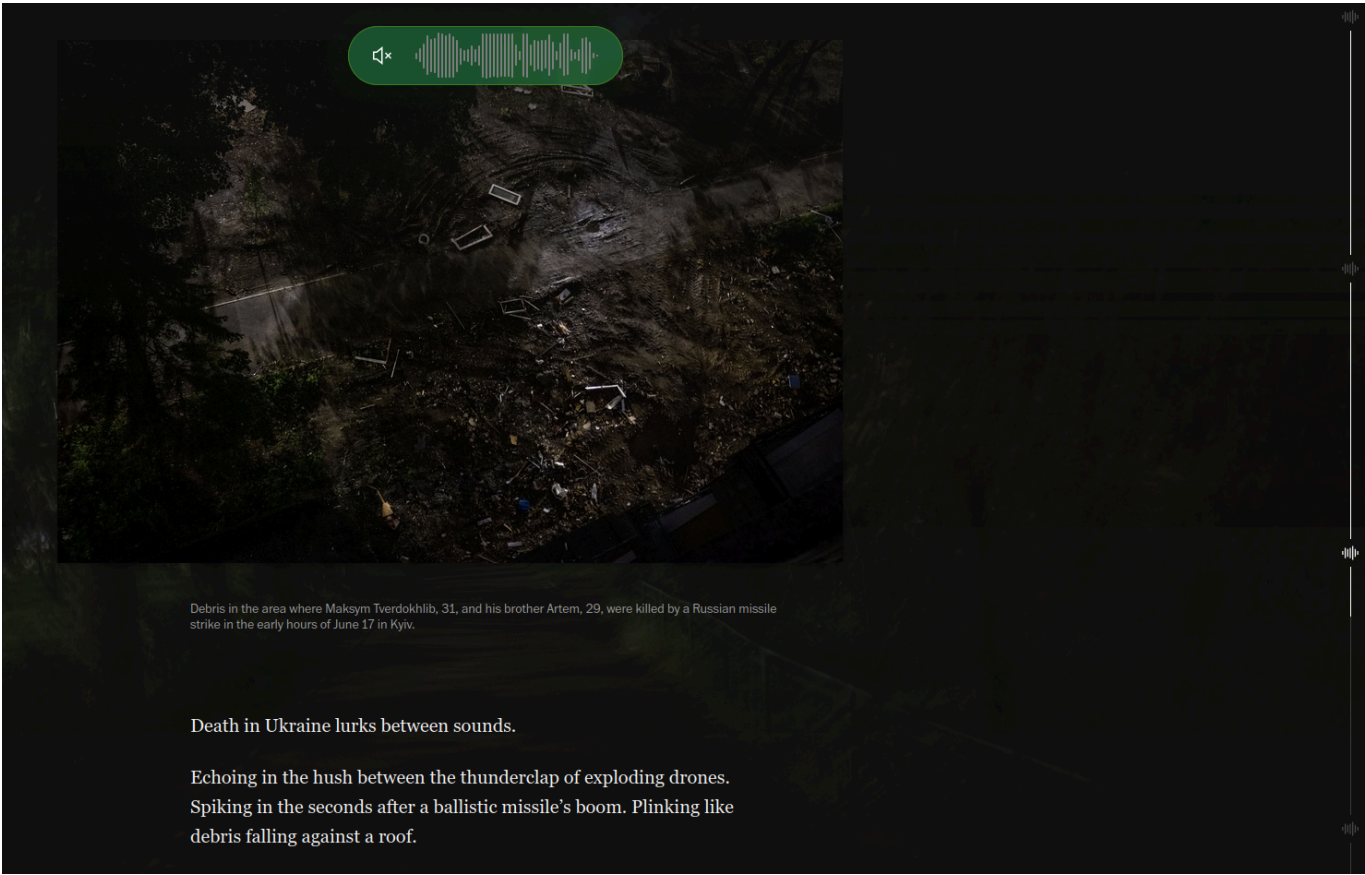
Context: Russian Bombings in Ukraine

By 2025, Russian drone and missile attacks had made sirens and explosions recurring sounds of life in Ukraine. Alone, displaying the amount of drones, missiles and alarm durations are harder to understand. This is why introducing noise is crucial to understand what repeated nighttime attacks sound and feel like.

The Washington Post combines data with recordings from the field and audio that plays as their readers scroll. Here, **sound is part of the evidence**, which places this example among the alternative approaches. Thanks to this the readers receive an immersive experience, by being able to both hear the environment of an attack and use the charts to verify how often attacks occurred.

Representations: Sound and Charts

The experience asks readers to turn on the sound and listen as they scroll.



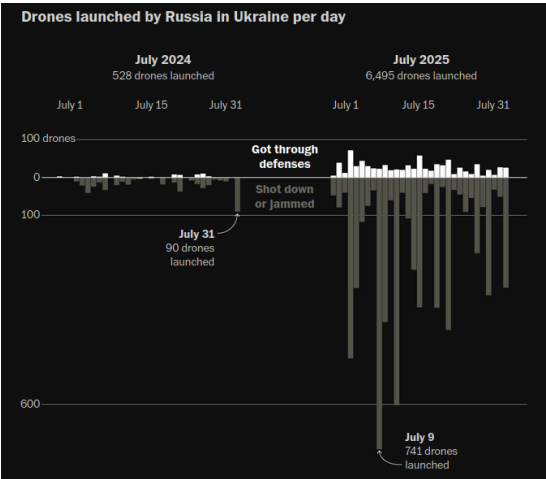
Recorded explosions, drones, and warning sirens accompany the story and graphics, including:

- Drone launches from January 2024 to July 2025;
- Daily report of drones launched, shot down, and jammed;
- Comparisons between the perceived loudness of war sounds and familiar everyday sounds;
- Timelines showing how many hours air-raid sirens lasted during the day and at night;
- Photographs, maps, and testimony that connect the sounds to the places where people live.

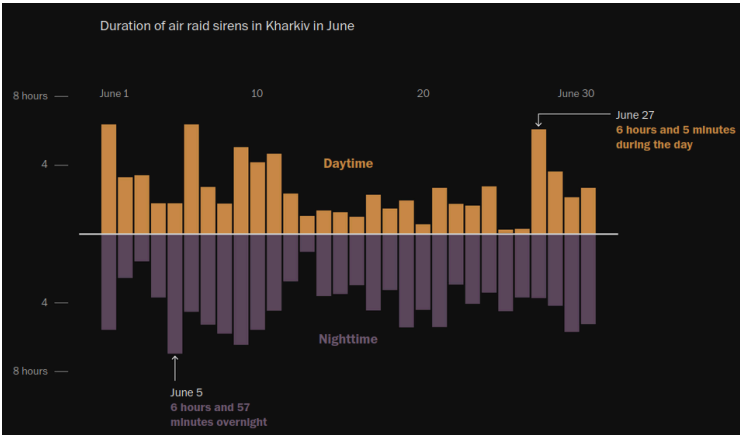
This is an **interactive sonic or aural data narrative**. It combines quantitative graphics with documentary recordings.

Here we look at two examples of charts:

Drones



Sirens



The way these charts are made seem to communicate a sense of "endlessness" with respect to the attacks and sirens.

The story moves between three scales:

1. Charts covering several months show the escalation;
2. Daily graphics distinguish launches from interceptions and jamming;
3. Sound and testimony bring readers back to one night and its duration.

Scrolling keeps these layers together, and an audio control lets readers choose when to listen.

Overall, this is made for an international audience on both desktop and mobile. While readers take part by scrolling and listening, they can use the charts to have a numerical comparison. The presence of the audio and images makes it easier for the viewer to empathize with Ukrainian civilians.

Knowledge: What can we learn from it?

The daily graphics separate "launched," "intercepted," and "reached the defended area," **helping readers distinguish the severity of an attack.**

The siren timelines show **how much time alarms take up**. This is important because people may spend hours uncertain whether an alarm will end quietly or with an impact. Here, showing both day and night differences helps us understand how this environment may cause multiple disruption to sleep and ordinary routines. As a consequence, this **helps the audience perceive** (through the concept of sleep), the **level of stress civilians live every day.**

Hearing the sounds adds something a table cannot provide. Loudness, repetitive sirens, and sudden explosions give a sense of distance and threat. Plus, pairing them with an empirical reference, keep the users grounded into the experience.

Taken together, the charts and recordings suggest that the attacks are becoming more frequent, occupying more time of the night, and repeatedly putting civilians on alert. This **helps readers understand the escalation as an experience of daily life.**

Pipeline from raw information to the visualizations

Unfortunately we have **no description** of how the visualization was made, just about the sources used.

Sources and tools

- **Primary source:** [The Washington Post interactive](#) and [Ukrainian armed forces](#) (for drone data), [Ukrainian authorities](#) (for air raid sirens data) and [World Health Organization](#) (for the perceived loudness of sounds)

Alternative Paradigm Representation 2

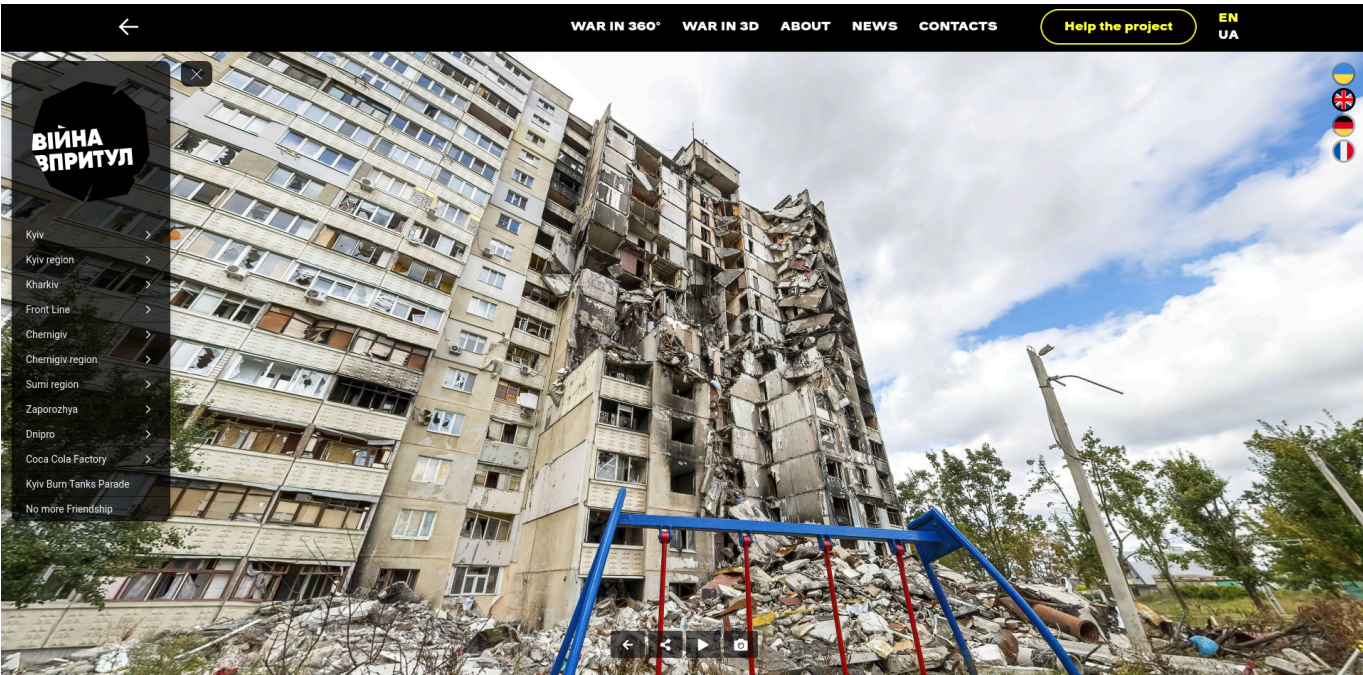
Title	Author(s)	Description	Link
War Up Close - VR Museum of the War in Ukraine	Mykola Omelchenko with Discover.ua, FreegenGroup, and project collaborators	A virtual-reality and immersive exhibition showing war damage in Ukrainian cities through 360-degree panoramas, drone footage, photogrammetry, and 3D models.	Official project

Context: Seeing the Damage of War

The project documents the physical effects of Russia's full-scale invasion of Ukraine: damage to residential buildings, infrastructure, streets, monuments, and public spaces. People can view the material online, through VR headsets, and in travelling exhibitions. It focuses on the view from inside damaged areas, with the belief that recording those views preserves evidence for memory and possible reconstruction.

Representations: Immersive VR and AR

First of all we encounter the street-view style of hotspots for destruction in Ukraine. The visualization app is embedded on the website as a "taste" of what the full experience can look like.



This **immersive spatial representation** allows the audience to put themselves around an interactive 360-degree panorama. What immediately catches the eye is the freedom that comes with this visualization: we are in the middle of a devastated area, usually inaccessible and dangerous, and we can look in different directions by moving our head or pointer to check/explore the state of the area. If this data remained in tabular format it would not be possible to translate the news of bombings into an almost palpable reality. This communication breaks the barrier of our war-blind reality and obliges the viewer to immediately recognize the destructive consequences of this conflict.

Another core element of the panoramas is represented by how rooms, streets, vehicles, debris, and neighbouring buildings connect. This allows the user to understand how the conflict has spread around these (mostly residential areas). Also, the absence of sound-related representations, adds a serious vibe to the experience.

The user can also switch between positions by changing hotspots or tour links.

There is also another kind of representation, given by **laser-scanned 3D models**.

Main / War in AR

Hotel "Ukraine", Chernihiv city

The "Ukraine" Hotel, with 90 rooms, was opened in the center of Chernihiv in 1961. For 60 years, it welcomed guests and was a symbol of the city.

On the night of March 12th, the Russians destroyed it by dropping a 500-kilogram aviation bomb FAB-500 onto the building. The roof and walls of the hotel collapsed, windows were blown out. According to architects' assessments, the structure is not suitable for restoration.

The work on 3D digitization was performed by the company [LibanCiel](#).



The website states that these are also used by the government to assess the amount of damage and plan for restoration.

Each view shows a different story, as we see damaged infrastructures, means of transport (cars, airplanes) and even military vehicles. This adds to the "ghost town" view.

Last but not least, we have to consider that the full experience is set in an exhibition. Therefore, using a headset we are able to make the experience more physical.

Knowledge: What can we learn from it?

Immersion can make the emotional impact stronger.

A table can record a location, coordinates, date, building type, and damage category. The immersive view shows **how the damage is arranged**. Viewers can see whether it affects one building or continues, how debris fills streets, and how damaged structures relate to surrounding infrastructures.

Moving between sky and street views helps with orientation and scale:

1. From above, viewers can take in an affected neighbourhood;
2. Within a panorama, they see its walls, rooms, and roads;
3. A label such as "residential building destroyed" becomes easier to understand as a place.

The **views also preserve details that a damage category leaves out**: explosion direction, exposed interiors, and which damaged/standing elements remain alongside one another. These observations can preserve testimony, help public understanding, and make assessments of the structures.

Overall, the project is mainly for the international public, including people far from Ukraine. It lets them look around documented places as active observers.

Pipeline from raw information to the visualizations

Unfortunately we initially had **no description** of how the visualization was made. However, the website states the following in their [About Us](#) section:

"The project is initiated by experts from Discover.ua and FreegenGroup. Mykola Omelchenko, a photojournalist with years of experience and the first Google Trusted Photographer in Ukraine, has embarked on photo shooting for the War up Close.

Using **state-of-the-art technologies, we capture the damage inflicted by the aggressor and put the photos on Google Maps, social media pages, and the project website**. For better immersing and a more realistic experience, **we show footage of Russian crimes with the help of VR goggles and immersive exhibitions**. Besides, a number of virtual museums of war has been created in Kyiv, Chernihiv, and Kharkiv regions in coordination with public agencies for tourism. [...]

The War up Close Project team is in close cooperation with the State Emergency Service, the Ministry of Internal Affairs, the Ministry of Culture and Information Policy of Ukraine and the Ministry of Foreign Affairs of Ukraine. Our **panoramic views and drone footages assist rescuers in debris removal and examining the actual scope of damage**. We also specialise in **3D modelling of monuments and historic buildings**.

Despite the dangers, we are capturing crimes against Ukraine and humanity for all the future generations to preserve the memory of this tragedy."

Then, we found this description at [The War up Close Project – memorialization of destructions](#):

"The Discover team have gone through several stages to create the VR museum:

1. Developing a detailed concept of the project, i.e. its general vision, idea and plan.
2. Obtaining permits for photo and video shooting from competent public authorities and agencies.
3. Conducting the shooting at great personal risk, as obtaining detailed footage often required us to come very close to destroyed and dangerous buildings.
4. Designing and programming the VR museum section.
5. Creating Google Street Views of the tours and adding them to Google Maps.
6. Spreading the word about the museum in mass media."

Sources and tools

- **Primary sources:** Documentary photographs, laser scans and videos taken at damaged sites in Ukraine.